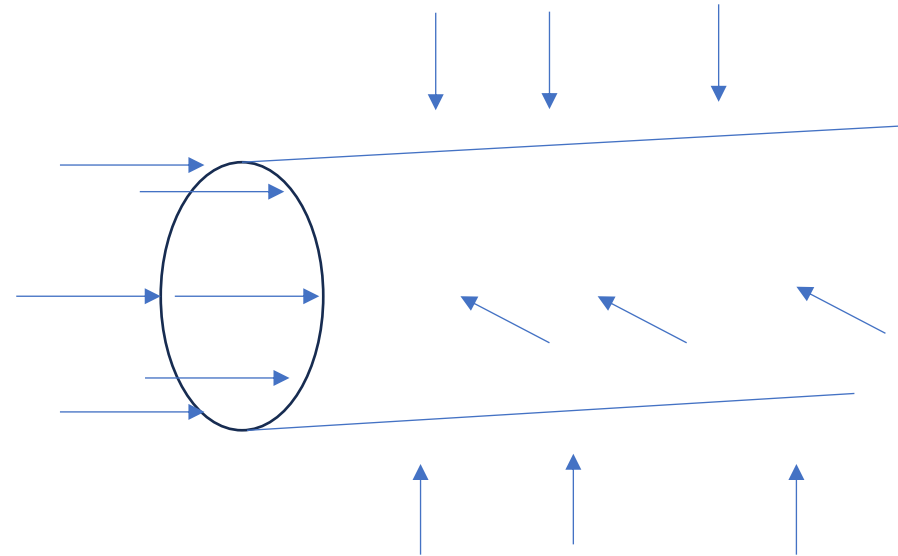


Ritz for Cylinders Under External Pressure

Plínio R. Santos

$$N_z = -\frac{p}{2R} - N_A$$

$$N_\theta = -\frac{p}{R}$$



Formulation

Ritz Method for **Stability Analysis** of Composite Cylinders

- Potential of the external load

$$V = -\frac{1}{2} \int_S N_z \left(\frac{\partial w}{\partial z} \right)^2 + N_\theta \left(\frac{\partial w}{\partial \theta} \right)^2 dS$$



$$\frac{\partial w}{\partial z} = [N_{w,z}] \{q_w\}$$

$$\frac{\partial w}{\partial \theta} = [N_{w,\theta}] \{q_w\}$$

$$V = -\frac{1}{2} \int_S N_z \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix}^T \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,z}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}^T \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix} + N_\theta \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix}^T \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,\theta}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}^T \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix} dS$$

$$W = -V = \frac{1}{2} \int_S N_z \{q\}^T [H_z]^T [H_z] \{q\} + N_\theta \{q\}^T [H_\theta]^T [H_\theta] \{q\} dS$$

$$W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] + N_\theta [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$[H_z] = \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,z}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

$$[H_\theta] = \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,\theta}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

$$N_z = -\frac{p}{2R} - N_A$$

$$N_\theta = -\frac{p}{R}$$

$$N_A = -\lambda \frac{p}{2R}$$

$$N_A = -\lambda \frac{N_\theta}{2}$$

$$\lambda = -\frac{N_A 2R}{p}$$

$$N_A = -\lambda \frac{p}{2R}$$

$$N_z = -\frac{p}{2R} + \lambda \frac{p}{2R} = \frac{p}{2R}(\lambda - 1) = -\frac{N_\theta}{2}(\lambda - 1) \quad N_\theta = -\frac{2N_z}{(\lambda - 1)}$$

$$W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] + N_\theta [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] - \frac{2N_z}{(\lambda - 1)} [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$W = \frac{1}{2} N_z \int_S \{q\}^T \left\{ [H_z]^T [H_z] - \frac{2}{(\lambda - 1)} [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

- In the equilibrium

$$\delta\pi_p = \delta U - \delta W = 0$$

$$\delta W = \frac{1}{2} \int_S N_z \{\delta q\}^T [H^*]^T [H^*] \{q\} dS + \frac{1}{2} N_z \int_S \{q\}^T \left\{ [H_z]^T [H_z] + 2[H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$\delta\pi = \{\delta q\}^T \left\{ \int_S [H]^T [C^*] [H] dS + N_z \int_S \{q\}^T \left\{ [H_z]^T [H_z] + 2[H_\theta]^T [H_\theta] \right\} \{q\} dS \right\} \{q\} = 0$$

$$[K_{Gz}] = \int_S \left\{ [H_z]^T [H_z] + 2[H_\theta]^T [H_\theta] \right\} dS$$

$$\{[K] + N_{z_{cr}} [K_{Gz}]\} \{q\} = \{0\}$$

$$N_{z_{cr}} = -\frac{p}{2R}$$

$$p = -2RN_{z_{cr}}$$

Cylinder buckling mode shape

